

RESEARCH INTERNSHIP PROJECT REPORT

DEVELOPMENT AND EVALUATION OF A
PHYSICS-INFORMED ML SURROGATE MODEL
FOR ROTARY ADSORPTION DESICCANT WHEELS

A Technical Research Project Report demonstrating the acceleration of coupled transient heat and mass transfer partial differential equations using degree-2 Polynomial Ridge Regression surrogates.

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Abstract

Solid desiccant wheels are crucial elements in advanced HVAC systems, providing deep air dehumidification necessary for industrial applications such as lithium-battery dry rooms (requiring Dew Point < -40°C), pharmaceutical formulation lines, and food storage. The physical modeling of these devices involves solving coupled, highly non-linear partial differential equations (PDEs) representing mass, momentum, and energy conservation inside sinusoidal honeycomb channels. Numerical finite-difference solvers suffer from severe computational latency, taking several seconds per run, which prevents real-time multi-variable sensitivity sweeps and cycle optimization.

This project introduces a hybrid Physics-Informed Machine Learning (PI-ML) surrogate solver. By generating a synthetic dataset of 5,000 simulations using a 1D spatial discretized solver, we trained a degree-2 Polynomial Ridge Regression surrogate to predict primary outlet states (temperature and humidity ratio). Exact physical conservation equations then derive secondary thermodynamic metrics (pressure drop, moisture removal rate, and thermal COP) in the client browser in under 0.02 milliseconds. The surrogate demonstrates an R² validation accuracy of 96.2% for temperature and 95.0% for humidity ratio. This report outlines the mathematical PDEs, sorption equilibrium, fluid dynamics, training pipeline, and the Large Language Model (LLM) advisor integration.

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1.0 Introduction & Background Literature

Commercial dehumidification is dominated by vapor-compression refrigeration, which cools air below its dewpoint to condense moisture. This approach is highly inefficient for dry rooms (dewpoints below -10°C to -50°C), as the cooling coils frost over, requiring energy-intensive defrost cycles. Solid desiccant dehumidification systems bypass this limitation by capturing moisture directly in the vapor phase using adsorbents (primarily silica gel or molecular sieves). The desiccant material is embedded on a rotary matrix, which continuously turns between process and regeneration air streams.

1.1 Industrial Applications & Requirements

Rotary desiccant systems are critical for maintaining low humidity in several key sectors:

- **Lithium-Battery Manufacturing:** Charging and assembly dry rooms must operate at a relative humidity under 1% (dewpoint under -40°C) to prevent lithium reacting with moisture to form toxic hydrofluoric acid.
- **Pharmaceuticals:** Solid dosage forms (effervescent tablets, powders) require strict moisture control below 20-30% RH during formulation to prevent caking and active ingredient degradation.
- **Agricultural Cold Chain:** Storing crops under controlled low humidity prevents mold development and increases shelf life without freezing the agricultural produce.

1.2 Computational Challenges & Limitations

Optimizing desiccant wheel parameters (thickness, channel height, RPM, air face velocities, and regeneration heater temperature) requires running multiple virtual experiments. However, the physical model involves non-linear coupled PDEs representing transient heat and mass transfer in porous media. Numerical solvers must discretize the channels spatially (e.g. 50 grid points) and temporally (e.g. 0.05s steps) over several wheel rotations to reach cyclic steady state. This takes 2 to 10 seconds of CPU processing time per simulation, making real-time sensitivity analysis or automatic control feedback impossible. An instantaneous, physics-informed surrogate model is therefore required.

2.0 Governing Physical Conservation Equations

The mathematical model represents a single channel of the honeycomb wheel. Process air flows through a channel of length L , transferring water vapor and heat to the desiccant coating on the channel walls. The equations assume one-dimensional flow, convective heat and mass transfer, and negligible axial conduction and diffusion.

2.1 Conservation of Mass (Water Vapor Balance)

The change in moisture concentration in the air flowing through the channel is directly balanced by the mass transfer rate into the desiccant coating:

$$\rho_a * v * (\partial Y / \partial x) + \rho_a * (\partial Y / \partial t) + \rho_d * ((1 - \epsilon) / \epsilon) * (\partial q / \partial t) = 0$$

Where ρ_a is the air density ($\sim 1.204 \text{ kg/m}^3$), v is the process air velocity (m/s), Y is the air humidity ratio (kg water/kg dry air), ρ_d is the dry desiccant material density (kg/m^3), ϵ is the channel porosity (void fraction), and q is the water content adsorbed inside the desiccant pores (kg water/kg dry desiccant).

2.2 Conservation of Energy (Enthalpy Balance)

The thermal energy balance accounts for sensible heating of the air stream and desiccant matrix, and the release of heat of adsorption as vapor molecules condense into the desiccant micro-pores:

$$\rho_a * v * (\partial h / \partial x) + \rho_a * (\partial h / \partial t) + \rho_d * ((1 - \epsilon) / \epsilon) * (\partial H_d / \partial t) = 0$$

Where h is moist air enthalpy ($h = C_{p,a} * T + Y * (C_{p,v} * T + h_{fg})$), H_d is the desiccant matrix enthalpy ($H_d = C_{p,d} * T_d + q * C_{p,w} * T_d - q * H_{ads}$), and H_{ads} is the latent heat of adsorption ($\sim 2800 \text{ kJ/kg water}$), which is higher than the heat of vaporization due to the electrostatic binding forces.

2.3 Interface Mass Transfer (Linear Driving Force)

The rate of moisture absorption from the air phase into the solid matrix is modeled using the Linear Driving Force (LDF) approximation:

$$\rho_d * (\partial q / \partial t) = k_m * a_v * (Y - Y_m(T_d, q))$$

Where k_m is the mass transfer coefficient (derived from the Sherwood number), a_v is the specific surface area (m^2/m^3), and Y_m is the equilibrium humidity ratio at the desiccant surface. This shows the coupled, non-linear nature of the equations: Y_m depends strongly on local temperature T_d and loading q .

3.0 Sorption Equilibrium & Channel Flow Dynamics

Solving the conservation equations requires defining the thermodynamic equilibrium relation (isotherm) of the desiccant adsorbent, and modeling the pressure loss of air flowing through the wheel matrix.

3.1 Desiccant Sorption Isotherm (Langmuir Model)

This project utilizes a silica gel (Type 1 desiccant) matrix. The equilibrium water content q_{eq} (kg water/kg dry desiccant) is calculated using relative humidity (ϕ) near the desiccant surface:

$$q_{eq} = q_{max} * (C * \phi) / (1 + (C - 1) * \phi)$$

Where $q_{max} = 0.40$ (maximum adsorption capacity) and $C = 2.5$ (kinetic energy constant). To evaluate the boundary humidity ratio Y_m , the local saturation pressure P_{sat} is calculated using the Antoine equation, and the vapor pressure P_v is derived:

$$P_{sat}(T) = \exp(23.196 - 3816.44 / (T + 227.02)) \text{ [Pa]} \quad P_v = \phi * P_{sat}(T) \quad Y_m = 0.622 * P_v / (P_{atm} - P_v)$$

3.2 Hydraulic Resistive Pressure Loss

To size process and regeneration fans, the air pressure drop across the sinusoidal channels of length L must be modeled. Since the flow inside the micro-channels is laminar ($Re < 1000$), the Fanning friction factor f is defined as $f = 16/Re$. This simplifies the pressure drop calculation:

$$\Delta P = f * (L / D_h) * (\rho * v^2) / 2 = (16.0 / Re) * (L / D_h) * (\rho * v^2) / 2 = 8 * \mu * L * v / D_h^2$$

Where μ is the dynamic air viscosity ($1.84e-5 \text{ Pa}\cdot\text{s}$), L is the wheel thickness (m), v is the face velocity (m/s), and D_h is the hydraulic diameter of the sinusoidal channels. For a sinusoidal shape, the hydraulic diameter is approximated as $D_h = 0.7 * h_{channel}$.

4.0 Machine Learning Surrogate Pipeline

To replace the iterative numerical solver, we built a hybrid surrogate model. Instead of predicting all outputs with a black-box model, the machine learning model predicts only the two primary independent thermodynamic states (outlet temperature T_{out} and outlet humidity ratio Y_{out}). All secondary variables are then derived using exact conservation equations.

4.1 Dataset & Feature Engineering

A synthetic dataset of 5,000 cases was generated using a 1D finite-difference physics simulation. The inputs cover the entire operational envelope. To capture the non-linear coupling of exothermicity, rotational speed, and geometry, the 9 input features are mapped to a degree-2 polynomial space:

Inputs: $x = [T_{in}, RH_{in}, v_{face}, RPM, T_{reg}, L, h, Flow_p, Flow_{reg}]$ Poly Expansion:
 $x_{poly} = [1, x_i, x_i * x_j, x_i^2]$ (Totaling 55 terms)

4.2 Ridge Regression Training & Browser Evaluation

A Ridge Regression model (L2 regularization parameter $\alpha = 1.0$) was trained in Python to identify the 55 coefficients for T_{out} and Y_{out} . The pre-computed weights were exported to the Vite/React application, executing predictions client-side using dot products under 0.02ms. The table below shows the performance metrics validated against the numerical solver test set:

Target Parameter	R ² Validation	RMSE	MAE
Outlet Temperature (T_{out})	0.962	0.82 °C	0.61 °C
Outlet Humidity Ratio (Y_{out})	0.950	0.34 g/kg	0.25 g/kg

5.0 Parametric Sensitivity & LLM Integration

The instantaneous execution of the surrogate model enables real-time multi-variable sensitivity sweeps and intelligent AI-driven diagnostics.

5.1 Operational Design Sensitivity

The model accurately captures key desiccant wheel design trade-offs:

- **Volumetric Flow vs. Dehumidification:** Higher air velocity increases volumetric capacity, but decreases contact time, resulting in a higher discharge humidity ratio and a quadratic increase in pressure drop.
- **Carryover Heat:** Higher regeneration temperatures dry the desiccant better, but transfer sensible heat directly into the wheel structure, raising process air outlet temperatures and lowering thermal COP.
- **Rotation Speed (RPM):** Running too slowly leads to desiccant saturation, while running too fast carries over sensible heat without adding moisture removal, defining an optimal RPM curve.

5.2 Large Language Model (LLM) API Integration

To assist engineers, the portal integrates an LLM API. The system feeds the user inputs, predicted states, and physical metrics into the API. The model evaluates the cycle against thermodynamic principles, identifies inefficiencies (such as thermal carryover or velocity bypass), and suggests optimization adjustments. This turns the tool from a passive calculator into an active design assistant.

5.3 Conclusions

The hybrid Physics-Informed ML Surrogate Model provides a fast, accurate method for rotary desiccant wheel simulation. By predicting primary states in 0.02ms with $R^2 > 95\%$ and deriving secondary metrics using conservation laws, it enables real-time parametric sweeps and optimization. Future work will explore expanding this surrogate to model multi-zone wheels and integrating Physics-Informed Neural Networks (PINNs).